K. J. Somaiya School of Engineering, Mumbai-77

Department of Sciences and Humanities

Course Name: Course Code:	Basic of Electrical Engineering BEEL) 316U06L102	Semester:	I
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Faculty Sign&Date:		Grade/Marks:	/

Experiment No: 4

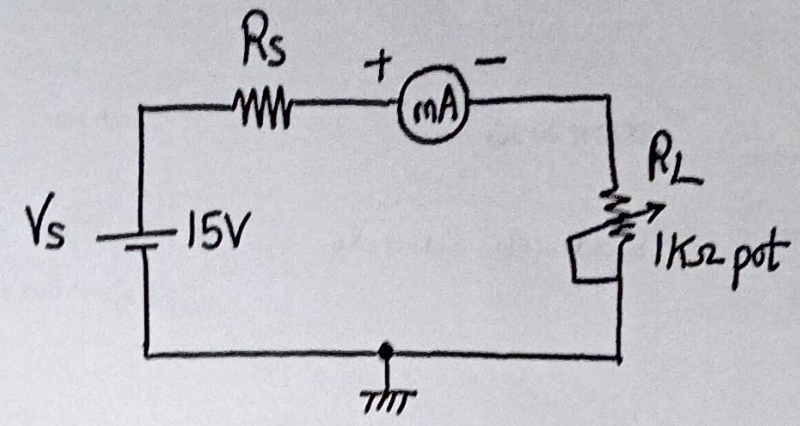
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Title: Maximum Power Transfer Theorem

Aim and Objective of the Experiment:
• To observe maximum power transfer across load resistor in a D.C circuit.

COs to be achieved:
CO1: Analyze resistive networks excited by DC sources using various network theorems.

Circuit Diagram:
V_s = 15 V and R_s = 560 Ω



Stepwise Procedure:

1. Set D.C. supply voltage V_S = 15 V
2. Vary R_L in the range 100 Ω - 1 KΩ in steps of 100 Ω
3. Note down I_L and V_L for each value of R_L. Where I_L and V_L are current through R_L and

   voltage across R_L respectively.
4. Prepare observation table showing readings of R_L Vs power $P = I_L \cdot V_L$
5. Plot graph of P Vs R_L

6. Locate the point of maximum value of power P and note down corresponding value of R_L.

Verify the results theoretically

Observation Table:

Sr. No.	$R_L\ \Omega$	Circuit Current (I_L) in mA		Voltage (V_L) in Volts	Power absorbed by load (P_L) in W $P_L = I_L^2 . R_L$	
		Theoretical	Practical		Theoretical	Practical
1.	100 108	22.7	22.8	2.1	51.5	44.1
2.	200 230	19.7	18.3	4.0	77.6	80.4
3.	300 309	17.4	16.1	5.16	90.8	88.3
4.	400 426	15.6	14.3	6.32	97.3	99.2
5.	500 511	14.1	14.1	7.09	100.1	100.5
6.	560 608	13.3	12.3	7.52	100.4	100.9
7.	700 704	12.93 11.9	12.1 10.1	7.78 8.26	99.51 99.12	100.8 97.3
8.	800 806	11.02	10.0	8.78	97.15	96.3
9.	900 932	10.27	9.8	9.2	95.3	94.8

10.	1K 1104	9.61	9.3	9.65	92.4	93.1

Graph: Draw a graph showing effect of variation in R_L on P_L using observation table. Take R_L on X – axis and P_L on Y- axis. (Use a graph paper)

This theorem states that the maximum power that can be transferred from source to load is 50%, which occurs when source impedance is matched.

Conclusion-

1. Explore one practical application where Maximum Power Transfer Theorem is used.
2. Draw a block diagram or circuit diagram of this application.
3. Explain in brief.

OR

Answer the following:

4. Do you apply Thevenin' s Theorem to calculate Maximum Power across load resistor in a D.C. circuit?
5. Take a sample problem. Draw a block diagram or circuit diagram of this sample problem.
6. Explain the solution in brief.

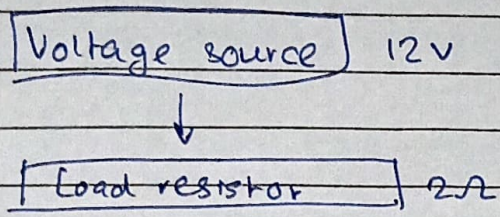
Signature of faculty in-charge with Date:

Q Do you apply Thevenin's Theorem to calculate Maximum Power across load resistor in a D.C. circuit?

→ Yes, Thevenin's Theorem can be used to calculate the maximum power across a load resistor in a D.C. circuit.

Q Take a sample problem. Draw a block diagram or circuit diagram of this sample problem.

→ Consider the following D.C. circuit:

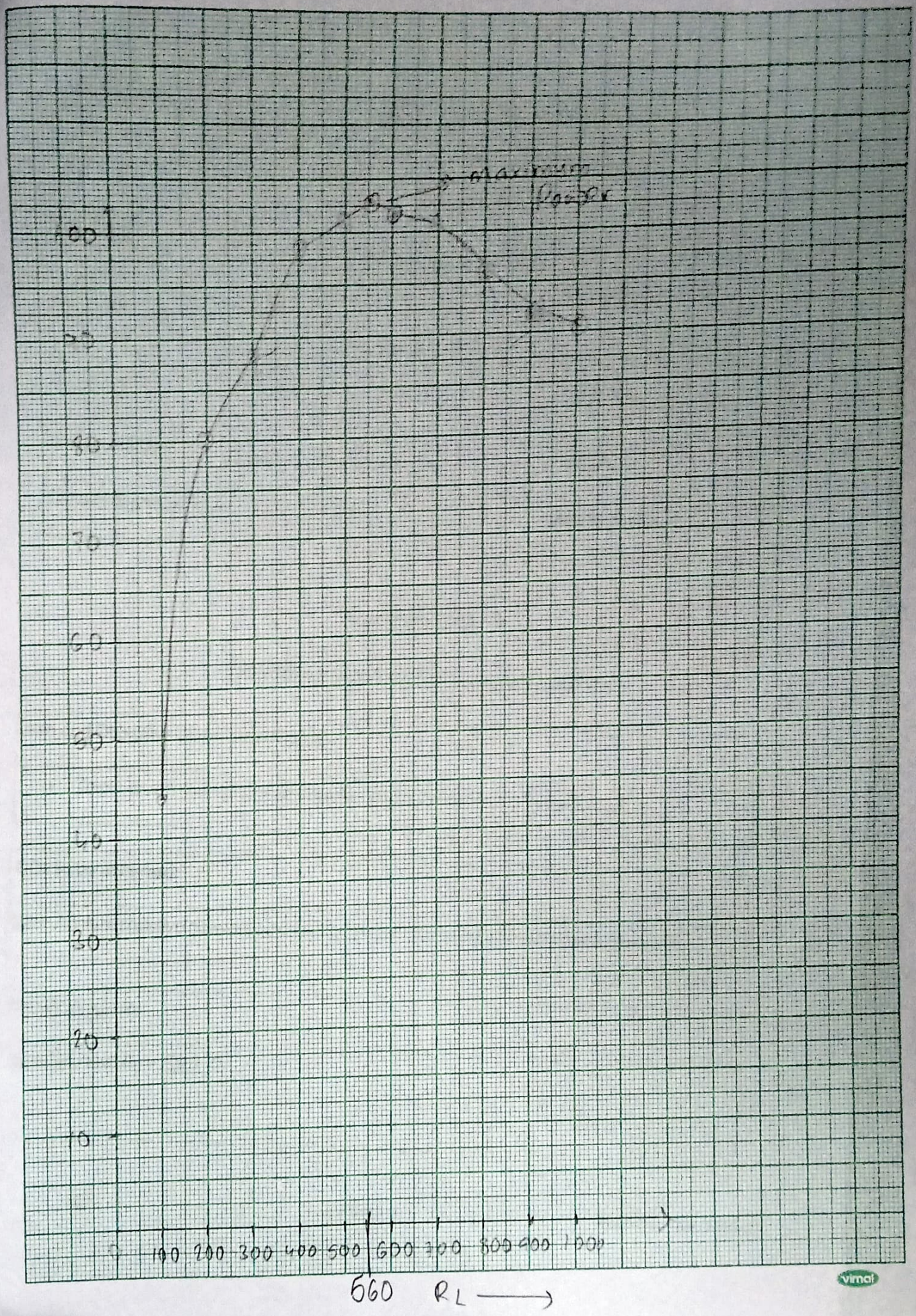


The voltage source has voltage of 12V and the load resistor has resistance of 2 ohms.

Q Explain the solution in brief.

→ The maximum power transfer in achieved when the load resistance is equal to the Thevenin resistance. In this case, the load resistance is already equal to the Thevenin resistance. Therefore, the maximum power that can be dissipated across the load resistor in the circuit is 96W.

Sunderam
FOR EDUCATIONAL USE


60
50
40
30
20
10
0
100 200 300 400 500 600 700 800 900 1000
560 R_L ⟶
R_L